Circles & Coordinate Geometry

ANSWER KEY

Section 1: General Circle Equations & Completing the Square

1 A circle has the equation $x^2 + y^2 + 4x - 10y + 13 = 0$. What are its center and radius?

a. **Center** $(-2, 5)$**, radius** 4 ✓

b. Center $(2, -5)$, radius 4

c. Center $(-2, 5)$, radius 16

d. Center $(2, -5)$, radius 16

Complete the square: $(x+2)^2 + (y-5)^2 = -13 + 4 + 25 = 16$. Center is $(-2, 5)$ and radius is $\sqrt{16} = 4$.

2 Find the equation of the circle that has points $A(1, 2)$ and $B(7, 10)$ as endpoints of a diameter.

a. $(x-4)^2 + (y-6)^2 = 25$ ✓

b. $(x-4)^2 + (y-6)^2 = 5$

c. $(x-3)^2 + (y-4)^2 = 25$

d. $(x+4)^2 + (y+6)^2 = 25$

Center is midpoint $(\frac{1+7}{2}, \frac{2+10}{2}) = (4, 6)$. Radius is distance from center to $(1, 2)$: $r = \sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9+16} = 5$. Equation: $(x-4)^2 + (y-6)^2 = 25$.

3 For what value of k does $x^2 + y^2 - 6x + 8y + k = 0$ represent a single point (degenerate circle)?

a. 25 ✓

b. 16

c. 9

d. 0

Completing square: $(x-3)^2 + (y+4)^2 = 9 + 16 - k = 25 - k$. For a single point, radius $= 0 \Rightarrow 25 - k = 0 \Rightarrow k = 25$.

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Section 2: Tangent Lines to Circles

4 What is the equation of the tangent line to the circle $x^2 + y^2 = 20$ at the point $(2, 4)$?

a. $x + 2y = 10$ ✓

b. $2x + y = 10$

c. $x - 2y = -6$

d. $2x - y = 0$

Radial line to $(2, 4)$ has slope $\frac{4}{2} = 2$. Tangent line has perpendicular slope $-\frac{1}{2}$:
$y - 4 = -\frac{1}{2}(x - 2) \Rightarrow 2y - 8 = -x + 2 \Rightarrow x + 2y = 10$.

5 Find the length of the tangent line drawn from external point $P(8, 0)$ to the circle $x^2 + y^2 = 16$.

a. $4\sqrt{3}$ ✓

b. 4

c. 8

d. $2\sqrt{5}$

The distance from center $(0, 0)$ to P is $d = 8$. Circle radius is $r = 4$. By Pythagorean theorem on the tangent right triangle: $L = \sqrt{d^2 - r^2} = \sqrt{8^2 - 4^2} = \sqrt{64 - 16} = \sqrt{48} = 4\sqrt{3}$.

6 At what points does the line $y = x + 2$ intersect the circle $x^2 + y^2 = 10$?

a. $(1, 3)$ **and** $(-3, -1)$ ✓

b. $(3, 1)$ and $(-1, -3)$

c. $(1, 3)$ only

d. No intersection

Substitute: $x^2 + (x + 2)^2 = 10 \Rightarrow 2x^2 + 4x - 6 = 0 \Rightarrow x^2 + 2x - 3 = 0 \Rightarrow (x + 3)(x - 1) = 0$. Points are $(1, 3)$ and $(-3, -1)$.

Section 3: Loci & Conic Definitions

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7 What is the locus of all points (x, y) that are equidistant from the point $(0, 3)$ and the line $y = -3$?

a. **Parabola** $y = \frac{1}{12}x^2$ ✓

b. Parabola $x = \frac{1}{12}y^2$

c. Circle $x^2 + y^2 = 9$

d. Ellipse $\frac{x^2}{9} + \frac{y^2}{6} = 1$

By definition, a parabola is the locus of points equidistant from focus $(0, p)$ and directrix $y = -p$. Here $p = 3$, so equation is $x^2 = 4py = 12y \Rightarrow y = \frac{1}{12}x^2$.

8 What is the locus of all points whose distances to $(2, 0)$ and $(-2, 0)$ have a constant sum of 6?

a. **Ellipse** $\frac{x^2}{9} + \frac{y^2}{5} = 1$ ✓

b. Hyperbola $\frac{x^2}{9} - \frac{y^2}{5} = 1$

c. Circle $x^2 + y^2 = 9$

d. Parabola $y^2 = 4x$

Constant sum $2a = 6 \Rightarrow a = 3$. Distance between foci $2c = 4 \Rightarrow c = 2$. $b^2 = a^2 - c^2 = 9 - 4 = 5$. Equation is ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$.

9 What geometric figure is defined by $x = 4 + 3\cos t$, $y = -1 + 3\sin t$?

a. **Circle centered at** $(4, -1)$ **with radius** 3 ✓

b. Ellipse with semimajor axis 4

c. Hyperbola centered at $(4, -1)$

d. Line with slope 1

Rearranging: $(x - 4)^2 + (y + 1)^2 = 9(\cos^2 t + \sin^2 t) = 9$. Circle centered at $(4, -1)$ with radius 3.

Section 4: Coordinate Proofs & Polygons

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10 What type of quadrilateral has vertices at $A(0,0), B(4,0), C(5,3),$ and $D(1,3)$?

a. **Parallelogram ✓**

b. Rhombus

c. Rectangle

d. Trapezoid (not parallelogram)

Slope of $\overline{AB}$ and $\overline{CD}$ is 0 (parallel, length 4). Slope of AD and BC is $\frac{3}{1} = 3$ (parallel). Length $AD = \sqrt{1^2 + 3^2} = \sqrt{10} \neq 4$. Thus it is a parallelogram.

11 Find the centroid of the triangle with vertices $(1,4), (5,-2),$ and $(6,7)$.

a. $(4,3)$ ✓

b. $(3,4)$

c. $(4,4)$

d. $(5,3)$

Centroid coordinates are averages: $(\frac{1+5+6}{3}, \frac{4-2+7}{3}) = (\frac{12}{3}, \frac{9}{3}) = (4,3)$.

12 What is the slope of the altitude drawn to side BC in $\triangle ABC$ where $B = (2,1)$ and $C = (6,5)$?

a. -1 ✓

b. 1

c. $\frac{1}{2}$

d. -2

Slope of $BC = \frac{5-1}{6-2} = 1$. The altitude is perpendicular to BC, so its slope is -1.

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